

Requirement Overview:

The key point is: **the client does not want an automation tool that depends on pre-existing test cases, flows, wireframes, or SMEs.** They expect the **QA Agent itself to learn the application from the URL and credentials**, understand the existing application/business behaviour, and then use that knowledge for future feature testing.

QA Agent – High-Level Client Requirements

1. AI-Based Application Discovery & Understanding

- The QA Agent should be able to explore the application using the provided **URL and valid credentials**.
- The agent should intelligently identify application modules, screens, navigation paths, user journeys, validations, and observable business behaviour.
- The solution should build and maintain an **application knowledge base** without requiring detailed functional documents, wireframes, or continuous SME involvement.

2. Business Flow Identification

- The agent should infer existing business flows and relationships from application exploration and available information.
- The client may provide only a **high-level business overview** rather than detailed test flows.
- The agent should progressively enhance its understanding as it explores and interacts with the application.

3. New Feature / Change Validation

- When a new feature or modification is introduced, the client should be able to provide a **basic feature document or description**.
- The QA Agent should understand the change in the context of the existing application.
- It should identify relevant existing flows and impacted areas.
- It should test the new or modified functionality and provide execution results.
- The solution should minimize the need for the client to manually create test cases or automation scripts for each change.

4. Natural Language Test Execution

- The QA Agent should accept **plain-English instructions** from users.
- Users should be able to instruct the agent to execute:
 - Sanity testing
 - Regression testing
 - Specific business flows
 - Feature-specific testing

- Exploratory validation
- The agent should interpret the instruction, identify the relevant flows, execute them, and provide the results.

Example:

"Run the sanity test for the application."

or

"Check whether a user can search for a product, view its details and complete the order flow."

The agent should understand the instruction and execute the appropriate flow.

5. On-Demand Automation Generation

- When requested, the QA Agent should generate automation scripts for flows that it has executed or identified.
- Automation generation should require **minimal manual scripting**.
- Generated scripts should be reusable for future regression or scheduled execution.

6. Self-Healing & Adaptive Automation

- The agent should identify changes in UI elements, locators, navigation, and supported application behaviour.
- Existing automation should be capable of **AI-assisted self-healing** when application changes occur.
- Where a change affects an existing flow, the agent should identify and adapt/regenerate the affected automation where technically feasible.

7. Change Impact Analysis

- The agent should identify which existing flows and automation assets may be impacted when a feature is added or modified.
- It should recommend or execute the relevant regression coverage based on the identified impact.

8. Automated Daily Sanity Testing

- The QA Agent should support **scheduled execution** of predefined sanity flows.
- A scheduler should allow the client to configure daily execution.
- The agent should automatically execute the configured sanity suite and generate the results.

9. Automated Reporting & Notifications

- Test execution results should be automatically generated after execution.
- Reports should include relevant execution status, failures, evidence, and other available insights.

- Reports should be distributed automatically through the **client's preferred notification channel**, such as email or an agreed reporting destination.

10. AI Test Analytics & Insights

- The agent should provide meaningful insights from test execution rather than only pass/fail results.
- Analytics should include, where supported:
 - Failed flows
 - Frequently failing areas
 - Regression trends
 - Automation coverage
 - Application changes affecting tests
 - Recurring failures
 - Recommended areas for investigation

11. Continuous Application Learning

- The QA Agent should maintain and progressively improve its understanding of the application.
- New discoveries, feature changes, business information, execution results, and approved automation should contribute to the application's knowledge context.
- The objective is to reduce the need to repeatedly teach the agent the same application behaviour.

The Core Requirement

I would put this **at the beginning of the requirements document**, because this is really what the Director is asking for:

The QA Agent should be capable of independently understanding and learning an application through application exploration using a provided URL and credentials, without depending on detailed functional documentation, wireframes, or continuous SME support. When new or modified functionality is introduced, the agent should use the existing application knowledge together with the available feature information to identify relevant test coverage, execute appropriate validations, generate automation when required, and provide actionable results. The agent should also support natural-language, on-demand test execution and scheduled daily sanity testing with automated reporting.

And there is one VERY important distinction

The client's expectation is **not simply**:

"Give me AI-generated automation."

It is:

URL + Credentials



Agent explores application



Agent learns application



Agent understands existing flows



New feature/basic document provided



Agent identifies what needs testing



Agent executes tests



Agent reports results



If required → generates automation



Future changes → detects impact / heals supported automation



Daily scheduler → executes sanity → sends report

That should become the **central architecture and product requirement** for QA Agent.

And importantly, this means your earlier "**client provides business flows / SME**" **dependency should no longer be presented as a mandatory prerequisite**. The client has explicitly said they will *not* provide that level of support. Their high-level overview becomes supplementary knowledge, while **application discovery must be a core capability of the agent**.